

Problem Statement

McDiffyStore is a vendor that sells a wide range of products across multiple categories. They have hired you to develop a **distributed e-commerce platform** to bring their business online.

As an initial MVP, you are required to design and implement a **database for a RESTful API backend application using Spring Boot**.

Requirements

1. Database Setup

- Use **H2 Database**.
 - All IDs must be **auto-increment (identity fields)**.
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2. Entities and Models

User

- `userId` – Integer
 - `username` – String
 - `password` – String
 - `roles` – List of Role
-

Role

- `roleId` – Integer
 - `role` – String
-

Category

- `categoryId` – Integer
 - `categoryName` – String
-

Product

- `productId` – Integer
- `productName` – String
- `price` – Double

- **seller** – User
 - **category** – Category
-

Cart

- **cartId** – Integer
 - **totalAmount** – Double
 - **user** – User
 - **cartProducts** – List of CartProduct
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CartProduct

- **cpId** – Integer
 - **cart** – Cart
 - **product** – Product
 - **quantity** – Integer
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UserRole (Mapping Table)

- **userId**
 - **roleId**
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3. Relationships

- A **User** can have multiple **Roles** (Many-to-Many).
 - A **User** can have one **Cart** (One-to-One).
 - A **Cart** can have multiple **CartProducts** (One-to-Many).
 - A **Product** belongs to one **Category** (Many-to-One).
 - A **Product** is sold by one **User (Seller)** (Many-to-One).
 - A **CartProduct** links **Cart** and **Product** (Many-to-One with both).
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4. Initial Data

Categories

categoryId	categoryName
1	Fashion
2	Electronics
3	Books

4	Groceries
5	Medicines

Roles

roleId	role
1	CONSUMER
2	SELLER

Users

userId	username	password
1	jack	pass_word
2	bob	pass_word
3	apple	pass_word
4	glaxo	pass_word

UserRole Mapping

userId	roleId
1	1
2	1
3	2
4	2

Cart

cartId	totalAmount	userId
1	20	1
2	0	2

Products

productId	productName	price	categoryId	sellerId
1	Apple iPad 10.2 8th Gen WiFi iOS Tablet	29190	2	3
2	Crocin pain relief tablet	10	5	4

CartProduct

cpId	cartId	productId	quantity
1	1	2	2

5. Final Task

- Implement all entities with proper **JPA relationships**.
- Ensure mappings and constraints are correctly defined.
- Refer to test cases for validation logic and expected behavior.